



Lighting up the quantum world: a dive into the photoelectric effect and Planck's constant

Logan Dales

*Department of Physics and Astronomy, Carthage College,
2001 Alford Park Drive, Kenosha, WI 53140, USA*

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The photoelectric effect caused classical physics and assumptions about energy to break down, resulting in the entire field of quantum physics to be created. In this report we use the photoelectric experiment set up to measure stopping voltages of different wavelengths of light in order to determine the value for Planck's universal constant from the slope of stopping voltage vs frequency plot. The importance of determining the value of a constant in different circumstances should never be squandered. Constants in physics are defined as fixed values and which have no uncertainty, so they must be true in any possible circumstance where they have an impact. Even though there will be some uncertainty in the value we find, the accepted value falls close to the middle of our range which is a good result for the accuracy of the experimental setup.

I. INTRODUCTION

The photoelectric effect caused classical physics and assumptions about energy to break down, resulting in the entire field of quantum physics to be created. In this report we use the photoelectric experiment set up to measure stopping voltages of different wavelengths of light in order to determine the value for Planck's universal constant from the slope of stopping voltage vs frequency plot. The importance of determining the value of a constant in different circumstances should never be squandered. Some constants physics, such as Planck's constant, are defined as fixed values and which have no uncertainty, so their precision determines the precision of the definition of SI units [1]. Even though there will be some uncertainty in the value we find, the accepted value falls close to the middle of our range which is a good result for the accuracy of the experimental setup.

German physicist Max Planck worked with black body radiation and discovered inconsistencies using predictions of the current theory at the time. This lead him in 1900 to create his theory that radiation, and subsequently energy, had to be in discrete packets [2]. "By defining that the energy had to be emitted in discrete packets, he had unknowingly postulated the existence of photons and ushered in the understanding of the quantum nature of energy and electromagnetic radiation" [3]. Later, Einstein took Planck's theory further to hypothesize that electromagnetic waves and thus light itself must also be quantize [4].

In classical theory, light was believed to be an electromagnetic wave with continuous energy. Think of the energy like a ramp, a person can choose to stand anywhere they want along it so there is an infinite number of possible positions. This idea was tested when light was shone on a photodiode surface where the energy from the

photons would be transferred to electrons in the material and they would gain enough energy to break their bonds and be ejected from the surface with a certain kinetic energy. The Photoelectric experiment has 3 results that cannot be explained by classical theory. First, there was no lag time from when the light was projected onto the surface to when the electrons were emitted; where classical thinking says there should be some form of build-up time as energy accumulates on the surface before having enough energy to break the electron's bonds. Second, there is no correlation between the intensity of light and the stopping voltage which is a measure of the maximum kinetic energy of the emitted electrons. Third, the existence of a minimum energy needed to start emitting electrons [4].

II. DETERMINING PLANCK'S CONSTANT

The experimental setup is simple: using a mercury lamp, shine filtered light onto an photodiode surface using a tunable DC (constant voltage) Power Supply and DC current amplifier and record the necessary voltage to stop current flow for each wavelength of light. To be transparent, our experiment setup has limitations (accuracy and possible effect of stray light entering the apparatus) and subsequent uncertainties greater than those of more exact setups with control over more variables and setups with more accuracy or data points. The results plotted in fig. 1 are for a 2mm diameter filter, but the results for the 4mm and 8mm were the same within the uncertainty determined for this experimental setup of ± 0.03 Amp range. This setup will be able to prove ways two and three of how classical expectations fail; to prove the first you would need a way to remove any standing charge and have an accurate timer to measure how long it takes for electrons to start current flow.

The explanation for the second revelation comes from Mr. Planck himself in the form of

$$E = hf, \quad (1)$$

which states a photon's energy (E) is only dependant on its frequency (f) and Planck's constant (h). This relationship of energy and frequency explains result 2 from above where intensity does not affect the stopping voltage. The photon-electron pairs are still going to individually have the same amount of energy to try and go upstream in the electric potential; intensity just means there will be more of them but they won't get any farther.

For the third one we need to calculate it from the graph of our data in fig. 1, but first some background knowledge. Stopping voltage is the electrical equivalent to the kinetic energy of the electrons being emitted from the photodiode material as energy is transferred from kinetic to electric potential energy seen as

$$\frac{1}{2}m\Delta v^2 = q\Delta V. \quad (2)$$

Where the mass (m), the charge (q), and the velocity (v) is relating to the electron and V is the voltage. The Δ 's are included in this case to emphasize the relationship of the velocity (KE) of the electron and how much voltage (PE) needs to be applied to stop it from reaching the cathode. This gives us two different energies that we could set equal to hf in eq. (1). we chose the kinetic energy one as this will tell us the maximum energy an electron has *after* it expends some amount of energy ϕ to break free of the material. This loss of energy is called the work function, is dependant on material, and is the answer for the third revelation. When you take that into account and plug in kinetic energy you are left with

$$K_{max} = hf - \phi. \quad (3)$$

Both Planck's constant and the work function can be found from looking at the results in fig. 1. h is related to the slope of the line of best fit. The units of the constant are Joule-seconds and the slope is in volt-seconds so in order for them to match you multiply volts by charge of an electron (1.6×10^{-19} Coulombs) to get potential energy units; also recall that you need to convert frequency back to base units as the x-axis is to the 10^{14} Hz. The result is:

$$h = (6.730 \pm .039) \times 10^{-34} \text{ Js.}$$

The work function ϕ can be found as the y-intercept on the graph, which is out of sight but can be found in the plot's legend as:

$$1.929 \pm .260 \text{ eV}$$

The work function is important to know for different materials because it is one factor that people should take into account when selecting a material for a special purpose like in semiconductors, electrical components, and

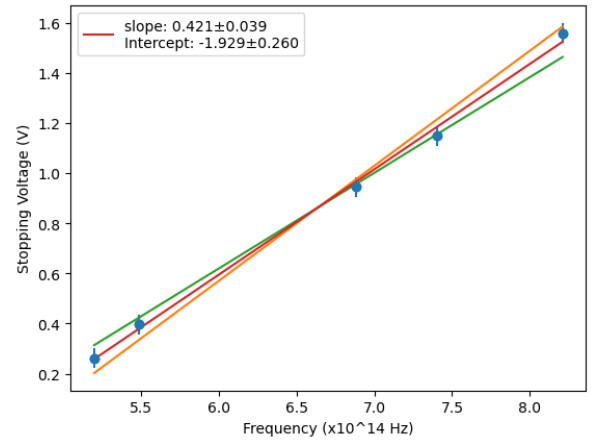


FIG. 1. This is a plot of the frequency of the light emitted vs the measured stopping voltage where the slope is related to the value of Planck's constant. The uncertainty of our measurement is taken into account by the vertical error bars and the orange maximum and green minimum slope variations while the red is the μ slope.

perhaps even in chemical reactions. While the value for Planck's constant is well known, findings agreeing values is always beneficial for the scientific community to be assured of its universal accuracy and precision when found using different methodologies. The accuracy of it is super important for quantum physics that uses Planck's constant in many of its calculations.

III. CONCLUSION

The photoelectric experiment confirms both Planck's and Einstein's theories about how energy is quantized into discrete packets in integers of Planck's constant (h). Planck's constant can be derived from the slope of stopping voltage vs frequency graph (fig. 1) that can be made from collecting simple data from the experimental setup, along with mathematically being able to determine the work function that Einstein theorized as the y-intercept of the same graph. Quantum physics was founded on the principals of defined energy quanta of photons and electromagnetic waves, and paved the way for modern physics to develop.

Experiments can often be improved or adapted to find better data or further test concepts. If you choose to take it upon yourself to find a more accurate value for Planck's constant you could either find a more accurate way to read the stopping voltage, or find more wavelength filters to get more than 5 data points. If we could change the experiment we would want to find out if adding in the Doppler effect would change the stopping voltage. Would it increase when moving the ends closer together and decrease when moving them apart, or would it have no effect at all. That could be for you to find out one day.

Appendix A: Uncertainty analysis

To determine the uncertainty in the value of Planck's constant we need to determine the uncertainty in the slope. First, to find the uncertainty in frequency (x-axis) we needed to find the standard deviation of the frequency:

$$\delta_f = \left\langle \sqrt{(f - \bar{f})^2} \right\rangle = 1.0328.$$

Next, the uncertainty we found in this experiment to be measurable was in finding the stopping voltage (y-axis). We estimated it was $\pm .04$ Volts which equated to the same current reading. we plugged both of those results into

$$\delta_m = \frac{\sigma_V}{\delta_f \sqrt{N}} = 0.039.$$

we only took one reading as opposed to doing the experiment several times and taking averages of the readings so N is simply going to be one here.

To find the uncertainty in the y-intercept b , or the work function ϕ , we needed to use some previously calculated values and plug them into the following:

$$\delta_b = \frac{\sigma_V}{\sqrt{N}} \sqrt{\left(1 + \frac{\bar{f}}{\delta_f}\right)^2} = 0.260.$$

Lastly to find the percent error between our experimental value of Planck's constant and the accepted value we did

$$\%error = \left| \frac{h_{experimental} - h_{accepted}}{h_{accepted}} \right| * 100 = 1.57\%$$

Appendix B: Acknowledgments

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